



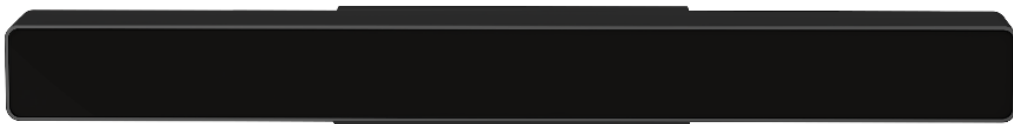
User Manual

Lite

LogicOne

v 1.1.16

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1 Welcome

EyeLogic's eye tracking devices meet the highest standards of accuracy and precision, required for scientific use. They are characterised by low and consistent latency, high robustness and high tolerance to head movement. What makes them unique, however, is the combination of quality criteria in connection with portability and ease of use.

EyeLogic currently offers two eye trackers: The LogicOne and the Lite. The LogicOne is a high precision remote eye tracking system capable of supporting frame rates up to 250 Hz. The Lite is designed for research both in and out of the lab. Its small size allows it to be used on smaller monitors and laptops and it supports frame rates up to 120 Hz.

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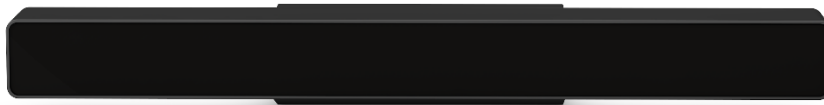
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2 System Overview

2.1 Product Overview

The LogicOne and the Lite are remote eye tracking devices for scientific use.



Product Name	LogicOne
Product Class	LD250
Sampling Rate	250 Hz, 120 Hz, 60 Hz
Gaze Accuracy	0.5°
Gaze Spatial Resolution	0.1°
Operational Distance	50 – 85 cm
Tracking Area	Width: 30 cm / height: 20 cm (@60cm)
PC interface	USB 3.1
Blink Recovery time	4ms (@250ms)
Latency	8ms (@250ms)
Technology	Dark pupil, non-invasive video based



Product Name	Lite
Product Class	LD120, LD60
Sampling Rate	120 Hz (LD120), 60 Hz
Gaze Accuracy	0.5°
Gaze Spatial Resolution	0.1°
Operational Distance	50 – 85 cm
Tracking Area	Width: 30 cm / height: 20 cm (@60cm)
PC interface	USB 3
Blink Recovery time	8ms (@120ms)
Latency	16ms (@120ms)
Technology	Dark pupil, non-invasive video based

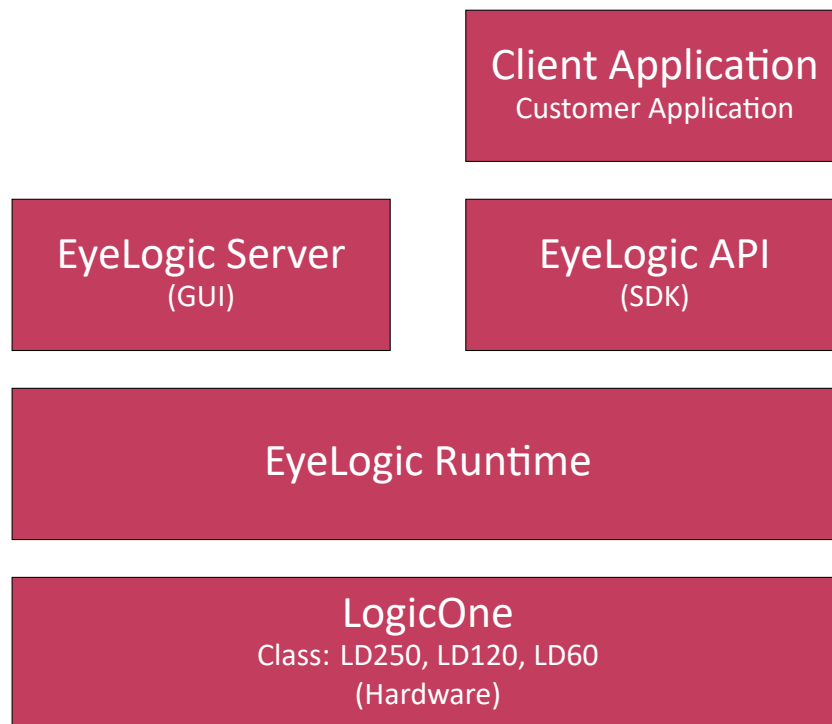
2.2 System Requirements

Recommended system specs are:

Operation System	Microsoft Windows 10
CPU	Intel(R) Core(TM) i7-7700HQ CPU @ 4 * 2.80GHz or higher
Disc Space	No special requirement, installation uses less than 100 MB
RAM	8 GB

2.3 Software Structure

The following diagram shows the system stack. On top of the LogicOne hardware, the runtime takes care of image acquisition, image processing and pupil and gaze calculation.



Once EyeLogic Server is initialised, a tray icon indicates the current status of the system.



A **dark gray** icon indicates the software is running and ready for use, with no accessible hardware device detected. The device may either be unconnected or unlicensed.



A **light gray** icon indicates the software is running, with a detected EyeLogic hardware device, Data generation is ready to be started, however there is no sample processing at this stage.

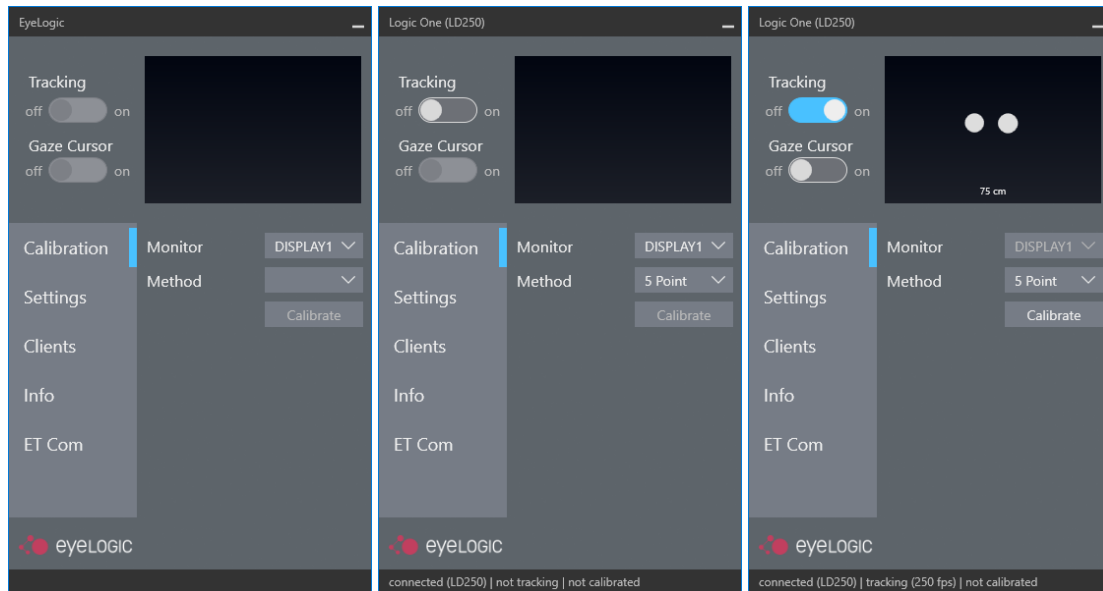


A **light blue** icon indicated the software is running, with a detected hardware device actively processing samples - i.e. the software is actively acquiring and processing images.

2.4 EyeLogic Server

This software runs as a background application, accessible via the tray icon, and contains the main functions: starting, setting up and calibrating the eye tracking system, as well as validating the quality of the calibrations. It includes a master switch (only enabled when a hardware device is detected), a visualisation of the calculated gaze (available after successful calibration), a tracking monitor to intuitively check the position relative to the device, and several menu options to customise the setup of the system.

The following images show the EyeLogic Server a) without a device connected, b) with a device connected but without tracking, and c) with tracking enabled.



Calibration is required to enable a tracking device to generate a valid data stream. Calibration adjusts the computation to produce an output stream optimized for the subject on which the calibration was performed. ATTENTION: This means that calibration must be performed each time a new subject is placed in front of the device. During a calibration, the subject is shown a series of stimulus points which are animated to move across the screen to a number of fixed positions where the points remain for a brief moment. The subject must focus on these points, particularly during the stationary periods. The software currently offers 1-point, 2-point, 5-point and 9-point calibrations. The 9-point calibration gives the best results but takes a little longer than the 1-point, 2-point or 5-point calibrations.

Validation can be used to check the quality of the last calibration. As such it is only available after a successful calibration. Procedurally it is equivalent to a 4-point calibration. However, the data is used to determine the mean deviation between the measured gaze direction and the expected ground truth for each of the 4 points. Validation is optional and does not need to be performed.

The **Calibration tab** provides the option to set the main display on which the gaze computation, calibration and validation will take place, a drop down menu to select the type of calibration required, the buttons to start both the calibration and the validation, and a feedback window to provide some information about the recent calibration or validation. The monitor selection drop down menu lists all the monitors connected to the system by their system given name. As changing monitors affects the system geometry, a new calibration must be performed each time a monitor is changed - again, only when tracking is disabled. After a successful calibration, the feedback window will indicate for each stimulus point whether it was successfully included in the calibration. After a successful validation, the feedback window shows for each stimulus point the mean deviation in degrees of each eye between the measured gaze direction and the expected ground truth.

The **settings tab** allows to:

- setup the monitor and correct its geometry, in case the automatic geometry estimation fails,
- setup the sampling rate, the possible choices of which are determined by the device class (LD250, LD120, LD60).

As the monitor, geometry and sampling rate affect the geometric model and temporal estimates, all settings can only be changed when tracking is disabled. If the monitor geometry auto-detection fails, manual correction is possible by unchecking the “Auto Detect” box and editing the respective properties. Note: These manual changes can significantly affect the quality of the gaze calculation.

Resolution is defined by the native pixel count on the selected monitor. Size is given in millimetres and is defined by the physical dimensions of the monitor matrix. Height is given in millimetres and is defined by the vertical difference between the top of the eye tracker and the bottom of the monitor matrix. Depth is defined as the horizontal distance between the top edge of the eye tracker and the monitor matrix. Angle (in degrees) is the upward tilt of the device (default is 20 degrees when mounted on the monitor using the standard mount).

The **Clients tab** lists all connected EyeLogic API Clients with name and tracking status. From here, each Client connection can be terminated individually. This version only allows one Client connection at a time.

Note: A Client can be remote, i.e. running on another computer connected to the actual computer via local network. If you wish to allow a Client to connect remotely to the actual EyeLogic Server, you must enable ‘Allow Remote Clients’ in the Clients tab.

The **Info tab** displays some general information (software version, serial number, etc.) and allows you to contact with EyeLogic by opening the default mail program.

The **ET Com tab** (only visible if the licence includes the SMI integration addon) is a licensable feature that enables support for the SMI iView X API ET command protocol. This protocol allows the EyeLogic eye tracking device to be controlled via Experiment Center and the iView X API.

As changes affect the tracking process, some changes are only allowed when tracking is disabled. When tracking is enabled, the user control fields for these options are disabled with a tooltip explaining why.

3 Getting Started

Preparation

Unpack the unit and attach the metal plate to the centre of the bottom edge of the display frame. A strip of double-sided adhesive tape is attached to the plate. You will need to remove the clear protective film before attaching it to the monitor frame. Do not attach it to the screen matrix itself, as this could damage the screen matrix.

Mounting

Mount the eye tracking device on the metal plate. The black Plexiglas cover must face the user, with the cable to the right of the monitor. The metal plate is exactly the same size as the cut-out on the back of the device. Built-in magnets hold the unit to the metal plate.

Installation

Run the **EyeLogic.msi** installer and grant admin rights if asked. Read the licence information carefully and accept if you understand and accept the licence agreement. If you do not have the installer, please contact EyeLogic Support.

Launching the software

The software can be accessed via the Start menu or via a desktop shortcut called “EyeLogicServer”. Once started, an EyeLogic **tray icon** will appear, giving you access to the **EyeLogic Server** software.

Plug in

Plug the eye tracker USB cable into a USB3 compliant USB port. After a few seconds, a Windows USB plugin sound will play. The EyeLogic Server tray icon indicates that the device has been detected and is ready for use.

Start Tracking

To open the GUI click on the tray icon: double click (left mouse button) or right click -> “Open Configuration”. The master switch for starting the tracking process is located at the top left of the window. As soon as it is pressed, the eye tracking device is activated (the infrared LEDs start to light up) and image processing begins. When a user is sitting in front of the device, the tracking monitor will show abstract information about his position in relation to the tracking box.

Setup Geometry

Disable tracking by toggling the master switch. Go to the **Settings tab**, which contains the display and the geometry settings. Normally these will be set automatically. However, if they are not correct, you can change them according to the geometry settings of your setup as described in the Geometry Settings Chapter.

Calibration

Enter the **Calibration tab** and select the preferred calibration mode. Have the user position themselves in the centre of the tracking box as indicated by the tracking monitor. Instruct the user to fixate on the calibration target throughout the process. Start the calibration process by clicking on the Calibrate button. To check the quality of the calibration, activate the gaze cursor by moving the “Gaze Cursor” slider to the right. This

slider is located at the top left of the window. This shows a real-time visualization of the calculated gaze point.

The calibration methods supported are 1-, 2- and 5-point calibration.

Validation - optional

Validation can be used after a successful calibration to provide an estimate of the quality of the calibration. To do this, the user is shown the calibration target again, which moves to 4 fixation points. The user is instructed to fixate the calibration target throughout the entire procedure. From the resulting data, the software calculates, for each eye and each fixation point, the mean deviation in degrees between the measured gaze direction and the expected ground truth.

The system is now calibrated and ready for use.

4 Product Classes

4.1 LD-Classes

The LogicOne is a remote eye tracking device for high accuracy, low latency research applications. Its configuration class (LD250) allows a maximum sampling rate of 250 Hz, so it can operate at 250Hz, 120Hz and 60Hz sampling rates, depending on the user's choice.

The Lite is a remote eye tracking device optimised for smaller monitors and laptops. It is configurable in two classes (LD120, LD60) that differ in their maximum sampling rate. The LD120 can run at either 120Hz or 60Hz depending on the user's choice. The LD60 operates at a sampling rate of 60Hz. All other quality criteria are the same.

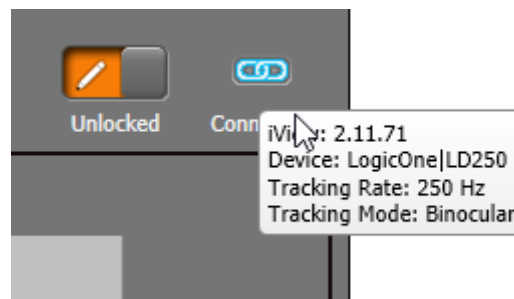
5 Additional Packages

5.1 Integration Addon: SMI Experiment Suite 360°

All EyeLogic eye trackers can be licensed with an additional integration addon for the “SMI Experiment Suite 360°” from SensoMotoric Instruments (SMI), which can be synchronised and accessed via SMI’s iView X API. Licensing this package allows the use of all iView X clients based on the latest iView X SDK version. This allows clients such as Experiment Center and BeGaze to configure, set up and use the EyeLogic system instead of an iView X remote eye tracking device. Although not all iView X API functions are currently supported, compatibility with SMI Experiment Center and SMI BeGaze is guaranteed.

5.2 Getting Started

To use Experiment Suite with an EyeLogic device, both software tools must be running. Start the tracking in the EyeLogic Server. Once Experiment Center is running and the eye tracker has started tracking, the connection button will change to indicate a successful connection. The tooltip will indicate the connected device, the tracking rate and the tracking mode.



From this point on, the Tracking Monitor displays the user’s position relative to the eye tracking device’s tracking box within the RED Tracking Monitor.



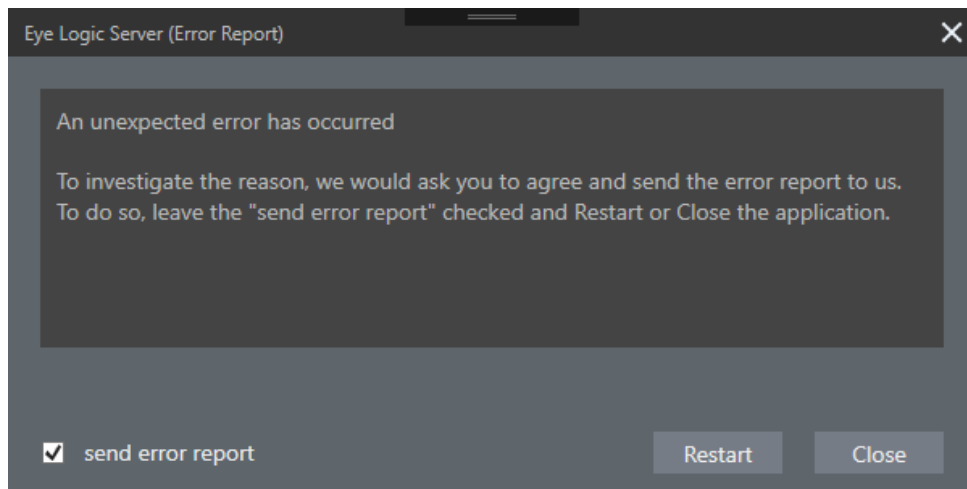
Although Experiment Center and the iView X API can be used to set up and maintain geometry settings, it is recommended that the Geometry tab in the EyeLogic Server is used to configure the geometry. In addition, Experiment Center is able to configure the calibration of the supported eye tracking devices using the Calibration Properties of Experiment Center. It is also possible to validate the current calibration from within Experiment Center. Due to the various types of eye tracking devices supported by Experiment Center, LogicOne/Lite does not support all of the features and settings listed in the Calibration Properties. In particular, the following features are handled differently when using the LogicOne/Lite:

- “Accept Calibration Point by” is always set to **Fixation (automatic)** by default and cannot be changed. The behaviour is the same as a calibration with the feature Smart Calibration feature enabled.
- Although “Calibration Method” is selectable for all different SMI devices (0 ...13-point), the supported calibration methods for LogicOne/Lite are limited to the calibration methods listed in the Calibration tab of the EyeLogic Server.

Once the eye tracker is connected to Experiment Center, all other process steps (experiment setup, experiment execution, calibration, validation, analysis) can be performed directly from Experiment Center and BeGaze.

6 Troubleshooting and Error Reporting

If errors occur, you will be notified with an error report dialogue. You can send the report and help us find the cause of the problem by keeping the checkbox checked while restarting/closing the notification. Close the window with , will not trigger the sending of a bug report, nor will it restart the software.



7 Compliance Information



LogicOne and Lite carry the CE mark, which certifies compliance with health, safety and environmental requirements for the European Union market.

LogicOne and Lite comply with the following safety standards:

IEC/EN 62471/2009 Photo Biological Safety of Lamps and Lamp Systems

2011/65/EU (RoHS 2) Restriction of Hazardous Substances Directive

2014/30/EU (EMC) Electromagnetic Compatibility Directive

LogicOne and Lite comply with the following EMC standards:

EN 55032/2015 Electromagnetic compatibility of multimedia equipment - Emission Requirements

EN 55035/2017 Electromagnetic compatibility of multimedia equipment - Immunity requirements

LogicOne and Lite comply with the following sustainability standards:

2012/19/EU (WEEE) Waste Electrical and Electronic Equipment Directive

8 Safety

8.1 Mounting

To ensure proper mounting of EyeLogic eye trackers, it is important to follow EyeLogic's instructions for approved mounts. The mounting bracket is versatile and can be attached to a variety of surfaces, especially single monitors. For safety reasons, it is not recommended to mount the eye tracker on monitors above the user's head or face, as it may fall. For the LogicOne, it is not recommended to mount it on a laptop screen.

8.2 Emergency Situations

EyeLogic eye trackers are intended for research purposes only. It is important to note that, although the risk is minimal, there is still a possibility of malfunction or distraction. It is therefore recommended that you do not rely on the eye tracker or use it in dangerous or critical situations.

8.3 Epilepsy

Certain people with photosensitive epilepsy may experience epileptic seizures or loss of consciousness when exposed to certain flashing lights or light patterns in their daily lives. This can happen even if the person has no history of epilepsy or previous seizures. People with photosensitive epilepsy may have seizures when looking at certain images or patterns on a screen or when exposed to the light sources of an eye tracker. Before a seizure occurs, many people with photosensitive epilepsy report experiencing an "aura" or unusual sensations. If you experience strange sensations while using an eye tracker, it is recommended that you move your eyes away from the device.

8.4 Infrared Light

Once activated, the EyeLogic eye tracker emits pulsed infrared (IR) light. Certain medical devices are susceptible to interference from IR light and/or radiation. To avoid compromising accuracy or functionality, it is not recommended to use the eye tracker near these types of sensitive medical devices. It is also important to avoid looking directly at the LED lights from a close distance (<15cm).

8.5 Magnetic Field

The EyeLogic eye tracker contains magnets that may interfere with the operation of cardiac pacemakers and implantable cardioverter defibrillators. As a general precaution, it is recommended that you keep a minimum distance of six inches (15 centimetres) between any object containing magnets and your cardiac device.

8.6 Child Safety

The EyeLogic eye tracker is a sophisticated electronic device and computer system made up of several parts. Some of these parts may come loose from the device in a child's hands, posing a potential choking or other safety hazard. It is strongly recommended that young children do not have access to the appliance or its accessories without appropriate supervision by a parent or guardian.

8.7 Electricity

Do not open the case of the EyeLogic eye tracker, as this may expose you to potentially harmful electrical voltages. The device does not contain any user serviceable components. Failure to comply will result in loss of warranty! If you experience any problems with the eye tracker, please contact the EyeLogic Support for assistance.

8.8 Accessories

Use only accessories supplied or approved by EyeLogic.

8.9 Third Party

Using an EyeLogic eye tracker in a manner other than its intended, or in conjunction with third party software or hardware that alters its intended function, is done at your own risk and EyeLogic GmbH cannot accept any responsibility in such situations.

9 Product Care

9.1 Operational, transportation and storage environment

The recommended temperature and humidity range during operation is:

Temperature: Min: 10°C Max: 35°C

Humidity: Min: 20% Max: 95% (no condensation)

The recommended temperature and humidity range during transport and storage is:

Temperature: Min: -20°C Max: 70°C

Humidity: Min: 20% Max: 95% (no condensation)

The LogicOne/Lite eye tracker is not waterproof or water resistant. The device should not be stored in excessively damp, wet or humid conditions. Do not immerse the eye tracker in water or other liquids. Take care not to spill any liquids on the device.

9.2 Cleaning

Before cleaning the EyeLogic eye tracker, disconnect the USB cable. To clean the front of the eye tracker, use a lens cleaning cloth and handle with care. Do not allow moisture to get inside the device. Do not use any other chemical cleaners such as window cleaners, household cleaners, aerosol sprays, solvents, alcohol, ammonia or abrasive cleaners. Scratches on the front of the eye tracker may affect its performance.

9.3 Disposal of the Eye Tracker

Do not dispose of your EyeLogic eye tracker in general household or office waste. Follow local regulations for the proper disposal of electrical and electronic equipment. We offer to take back your old device, please contact us for more information.

9.4 Limitation of liability

EyeLogic accepts no liability for damage to property or personal injury resulting from misuse of the product or damage caused by improper use or failure to follow the instructions in this manual. In addition, the warranty is void if the product is modified or repaired by unauthorised persons.

10 License Agreement

IMPORTANT - PLEASE READ CAREFULLY:

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 - (b) The application or third party content and services will be uninterrupted, accurate, reliable, timely, secure or error-free
 - (c) The quality of any products, services, information or other material accessed or obtained by you through the application will be as represented or meet your expectations, or
 - (d) Any errors in the application or third party content and services will be corrected.

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 - (b) was already known to licensee prior to the disclosure by EyeLogic, or
 - (c) was rightfully acquired by licensee from a third party without a breach of a confidentiality obligation towards EyeLogic.

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11 About EyeLogic

EyeLogic is a manufacturer of high precision and high quality eye tracking devices, mainly for scientific and research use cases. EyeLogic GmbH is a spin-off of the Free University of Berlin, faculty of mathematics and computer science and has a vast experience in image processing and computer vision.

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